Proof. By Proposition 3.5.8, (i) implies (ii), and to prove (i), it is sufficient to show that if V is a finite dimensional representation of G and $W \subset V$ is any subrepresentation, then there exists a subrepresentation $W' \subset V$ such that $V = W \oplus W'$ as representations.

Choose any complement $\widehat{W}$ of W in V. (Thus $V = W \oplus \widehat{W}$ as *vector spaces*, but not necessarily as *representations*.) Let P be the projection along $\widehat{W}$ onto W, i.e., the operator on V defined by $P|_W = \mathrm{Id}$ and $P|_{\widehat{W}} = 0$. Let

$$\overline{P} := \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g^{-1}),$$

where $\rho(g)$ is the action of g on V, and let

$$W' = \ker \overline{P}.$$

Now $\overline{P}|_W = \mathrm{Id}$ and $\overline{P}(V) \subseteq W$, so $\overline{P}^2 = \overline{P}$, and so $\overline{P}$ is a projection along W'. Thus, $V = W \oplus W'$ as vector spaces.

Moreover, for any $h \in G$ and any $y \in W'$,

$$\begin{aligned} \overline{P}\rho(h)y &= \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g^{-1}h) y \\ &= \frac{1}{|G|} \sum_{\ell \in G} \rho(h\ell) P \rho(\ell^{-1}) y = \rho(h)\overline{P}y = 0, \end{aligned}$$

so $\rho(h)y \in \ker \overline{P} = W'$. Thus, W' is invariant under the action of G and is therefore a subrepresentation of V. Thus, $V = W \oplus W'$ is the desired decomposition into subrepresentations. $\square$

The converse to Theorem 4.1.1(i) also holds.

Proposition 4.1.2. *If $k[G]$ is semisimple, then the characteristic of k does not divide $|G|$.*

Proof. Write $k[G] = \bigoplus_{i=1}^r \mathrm{End}\ V_i$, where the V_i are irreducible representations and $V_1 = k$ is the trivial 1-dimensional representation. Then

$$k[G] = k \oplus \bigoplus_{i=2}^r \mathrm{End}\ V_i = k \oplus \bigoplus_{i=2}^r d_i V_i,$$